Based on the Winning Concept Cultivated at Climathon Lappeenranta 2024:

Mobile Application to Track and Reduce Digital Carbon Footprint

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Abstract

The increasing digitalization of society contributes significantly to carbon emissions through energy-intensive data centres, network infrastructure, and device usage. However, limited accessible data on the digital carbon footprint makes it challenging for individuals to understand the environmental consequences of everyday digital activities—such as browsing social media, streaming videos, or using AI tools like ChatGPT.

This paper presents a mobile application developed during Climathon Lappeenranta 2024, which addresses the challenge of reducing digital carbon emissions. The winning concept was a mobile application that tracks users' screen time and calculates their carbon footprint based on their device usage. By combining intuitive design with educational features, the app visualizes emissions data, provides motivational feedback, and bridges the gap between users and actionable insights, enabling individuals to adopt more sustainable digital habits.

The methodology relies on Greenspector's 2021 report to calculate emissions, supplemented by plans for more comprehensive datasets and future iterations. Initial evaluations based on feedback from Climathon participants indicate significant potential for the app to raise awareness, influence behaviour, and contribute to the growing body of research on digital sustainability. While the beta version highlights the feasibility of using gamification and user feedback to promote environmental responsibility, future work will expand accessibility, enhance data accuracy, and integrate innovative features to maximize the app's impact and scientific relevance.

CCS Concepts

- Human-centered computing → Mobile applications
- Computing methodologies → Sustainability
- Applied computing → Environmental sciences

Keywords

Digital carbon footprint, sustainability, mobile application, user engagement, carbon tracking, gamification, sustainable software engineering

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1. Introduction

Digital technologies are integral to modern life, shaping how people communicate, work, and entertain themselves. However, the environmental impact of these digital actions is often overlooked. Internet usage, video streaming, and app interactions require significant amount of energy, much of which is consumed by data centres, network infrastructure, and personal devices. This energy use indirectly contributes to greenhouse gas emissions, yet many individuals remain unaware that everyday digital activities, such as meticulously prompting an AI tool to write your History paper or doomscrolling TikTok for six hours, have as tangible environmental consequences as for example driving a car.

One promising avenue for addressing this knowledge gap is through mobile applications. Mobile apps are highly accessible and widely used, making them an effective platform for disseminating information and fostering behavioural change. By gamifying learning experiences and integrating agile features, mobile applications provide an engaging and interactive way for users to understand complex topics, such as the digital carbon footprint. This application approach has the potential to visualize the often-invisible impacts of digital habits and motivate users to adopt more sustainable practices through real-time feedback and actionable insights.

This paper details the winning concept of Climathon Lappeenranta 2024: a mobile application designed to track and reduce digital carbon emissions. The application calculates emissions based on screen time and app usage, visualizing the resulting data to raise awareness. By combining user-friendly design, gamified elements, and motivational features, the app empowers individuals to make more informed and environmentally responsible choices in their digital lives. Through this approach, the concept aims not only to reduce emissions but also to bridge the gap in understanding how digital habits impact the environment.

2. Related Work

The concept of the digital carbon footprint is gaining attention as society becomes increasingly aware of the environmental impact of digital technologies. While the body of literature on this topic is still relatively young, several studies have begun to explore its implications across different domains, highlighting the challenges and opportunities for fostering sustainability in the digital age.

De Moore et al., 2021 explored digital natives' awareness of their digital carbon footprint through interviews. They found low awareness due to limited public information and technological understanding. Personal well-being often motivated eco-friendly behaviours, but willingness to change depended on perceived sacrifices. The study calls for better awareness tools and sustainable design practices.

Bhatnagar et al., 2024 focus on reducing the carbon footprint of virtual worlds by exploring strategies such as hardware optimization, software efficiency improvements, and renewable energy integration. The study highlights the environmental impacts of data centres and digital ecosystems and discusses the socio-economic and policy implications of reducing emissions. It emphasizes collaborative initiatives, technological innovations, and public awareness campaigns to foster sustainable practices in virtual environments.

Leila Pourghannad's (2024) thesis examines the role of social media in reducing the digital carbon footprint by educating users on sustainable consumption. Utilizing the Theory of Planned Behaviour, the research identifies low awareness among users about their digital carbon footprint and its environmental impact. Through surveys and interviews, the study finds that education and awareness campaigns can motivate eco-friendly behaviours, such as mindful platform usage and content sharing.

It recommends time management tools, sustainability campaigns, and partnerships between platforms and environmental organizations to foster sustainable practices online.

Dudukalov et al., 2024 investigate the environmental impact of digital banking in Asia, emphasizing its role in reducing carbon footprints. Through Structural Equation Modelling (SEM) and data from 520 respondents, the study finds that adopting digital banking services and increasing digital payments significantly reduce paper usage and energy consumption. Customer awareness of green banking features is pivotal for fostering eco-conscious behaviours. The research highlights the importance of digital transformation, paperless practices, and customer education in achieving sustainability goals within the financial sector.

3. Methodology

This study focuses on the development of a mobile application to track and visualize users' digital carbon footprints, promoting sustainable digital habits. The methodology includes the design and functionality of the application, the data sources used for carbon footprint calculations, and the features implemented to encourage user engagement.

3.1 Application Structure

Upon launching the app, users are presented with a view titled *"Digital Carbon Footprint Today"*, which displays their daily carbon footprint in kilograms of CO2 (Figure 1).

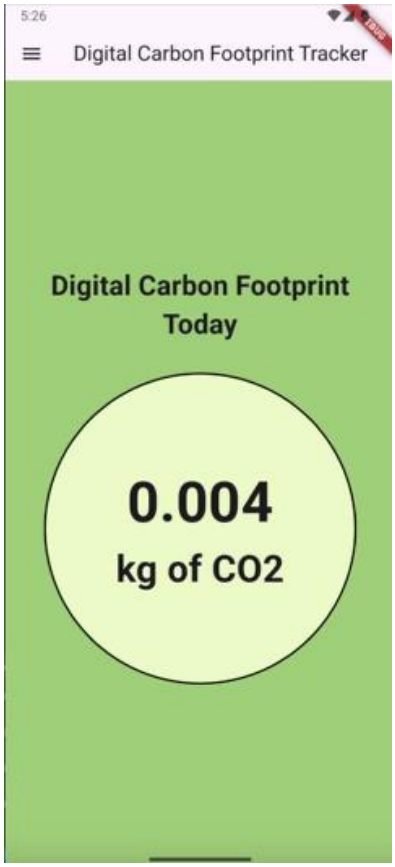

Figure 1: Main interface

These values are calculated based on emission metrics provided in Greenspector's 2021 report, which quantifies the energy consumption and emissions of common digital activities and applications. The main interface of the application also includes a hamburger menu on the top-left corner, allowing users to navigate to the following sections: Home, Usage Statistics, Total Impact, and About (Figure 2).

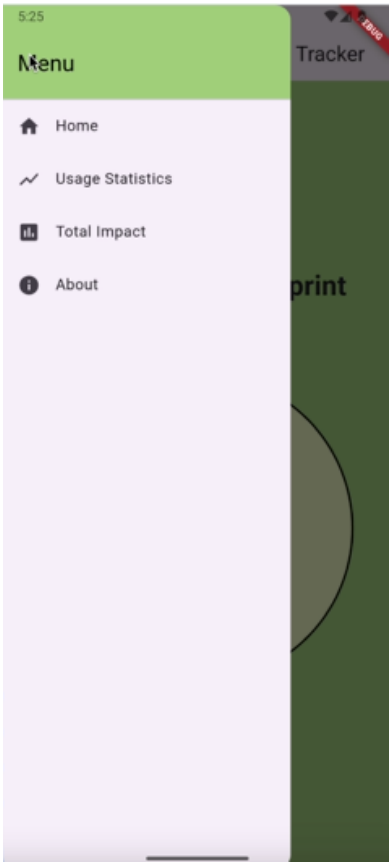

Figure 2: Menu

The design emphasizes clear structure and base features combined with user-friendly navigation.

3.2 Carbon Footprint Calculations

The calculation of emissions relies on Greenspector's 2021 report, which offers a detailed analysis of energy usage and carbon emissions for popular applications. These numbers are integrated into the application's backend to estimate the CO2 emissions based on the user's screen time and app usage. While current calculations use generalized data for each application, future iterations aim to include more precise, device-specific data.

3.3 Features and Functionality

The application includes the following key features:

- **Usage Statistics:**
  This section (Figure 3) visualizes the user's emissions through a donut chart, categorized by different applications they have used, such as TikTok, Twitch, Snapchat, X, Facebook, Pinterest, YouTube, Instagram, LinkedIn, and Reddit. Users can view their emissions data on a weekly, monthly, or yearly basis by toggling options within this interface. This visual breakdown helps users identify which applications contribute most to their digital carbon footprint.

- **Total Impact:**
  This section will in the future provide a longitudinal view of the user's emissions, comparing changes over previous weeks, months, or years when the necessary data is available. Emissions are presented in kilograms of CO2 (Figure 4).

- **About:**
  The About section serves as a disclaimer, explaining the app's purpose, the methods used for calculating emissions, and the limitations of the data (Figure 5).

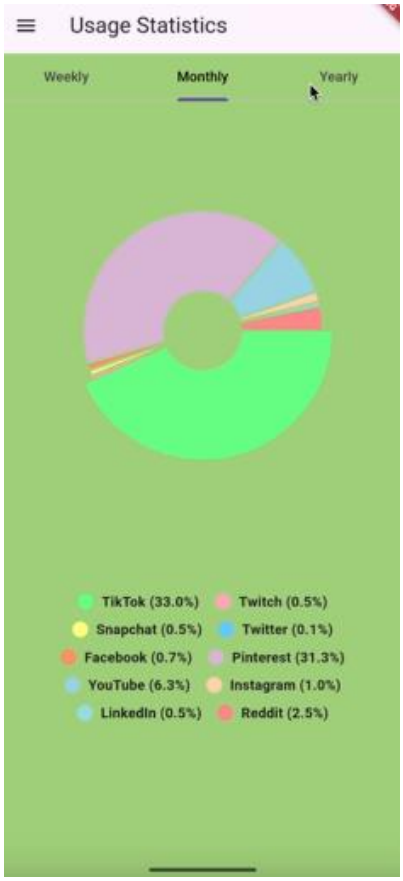

Figure 3: Usage Statistics

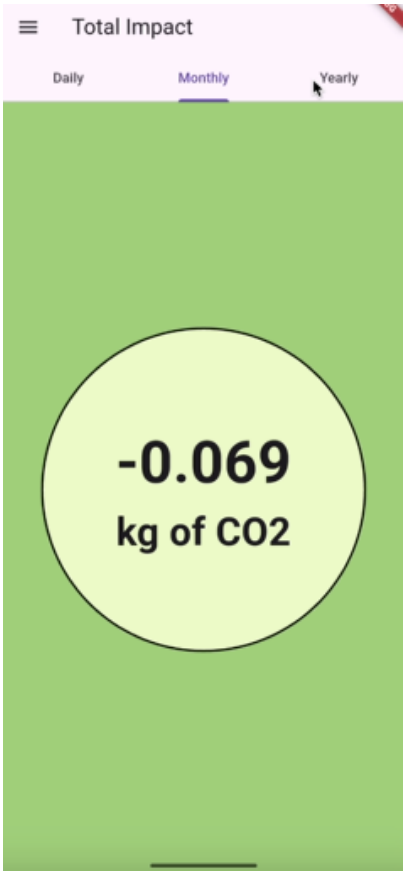

Figure 4: Total Impact

Figure 5: About

In future updates, the application will include push notifications to keep users informed about their carbon footprint. Additionally, a new feature will allow users to set a target emission level they aim to stay within. The application will send notifications to update users on their progress toward this target, encouraging them to maintain more sustainable digital habits.

3.4 User Interface and Experience

The application is designed with a strong emphasis on simplicity and usability to ensure that users can easily navigate and engage with its features. Key interface elements, such as donut charts for visualizing emissions and summary views for tracking progress, are deliberately designed to be visually engaging and intuitive. These features provide users with clear, actionable insights into their digital habits.

In the future, the application will provide comparisons between the user's emissions (measured in kilograms of CO2) and tangible physical activities to help demonstrate the environmental impact of their behaviour. For instance, users will be able to see how their emissions correspond to the equivalent amount of beef produced or the distance travelled in kilometres, offering a more relatable and concrete understanding of their digital carbon footprint.

By maintaining a balance between simplicity and functionality, the application seeks to create an effective tool that is both engaging and purpose-driven, making it easier for users to understand and reduce their digital carbon footprint.

To avoid overwhelming users, the application focuses on a few core features that align with its primary goal: educating users about their digital carbon footprint and encouraging sustainable digital habits. By prioritizing essential functionalities, such as emissions tracking, progress monitoring, and basic visualizations, the app avoids cluttering the user experience with unnecessary gadgets or features that may distract from its core purpose.

Currently, the beta version of the application does not include advanced accessibility features, such as support for multiple languages, customizable color schemes, light/dark mode, text-to-speech functionality, or adjustable text sizes. While these features are not essential for testing the concept, their absence may limit the app's inclusivity and usability for diverse audiences. Future iterations of the application aim to address these limitations to ensure a more accessible and user-friendly experience for a broader range of users.

3.5 Technical Details

The application was developed using the Dart programming language, selected for its versatility in building cross-platform mobile applications and its suitability for developers with limited prior experience in mobile development. Before starting the project, research was conducted into various tools and frameworks available for mobile application development. This research was essential, as we were not initially familiar with mobile development processes or technologies.

Dart stood out due to its dynamic nature and simplicity, offering a gentle learning curve compared to alternatives like JavaScript-based frameworks, which can involve a steeper learning process and more complex configurations. Dart’s integration with the Flutter framework further enhanced its appeal, providing a robust toolkit for creating visually appealing, responsive, and performant applications for both Android and iOS platforms from a single codebase. These factors made Dart an ideal choice for rapidly developing a functional prototype while allowing the team to focus on the core features and usability of the application rather than overcoming technical barriers.

By adopting Dart, it was possible to efficiently translate the concept from Climathon Lappeenranta into a fully realized mobile application, overcoming initial knowledge gaps and leveraging a development environment that balanced flexibility and ease of use.

3.6 Testing and Validation

The app's development process included initial testing focused primarily on navigation and the basic structure of the application. These aspects were discussed with the students tasked with implementing the concept and the professors overseeing the project.

The idea originated from the winning team of Climathon Lappeenranta 2024, who conceptualized the app as an educational tool. With their agreement, the concept was developed into a concrete application. However, broader testing or public release has not yet been conducted, as further steps require formal approval from the concept owners to ensure alignment with their original vision.

The first user data was collected on the developer's device and the initial results show that the application successfully illustrates user's digital carbon footprint divided into different applications (Figure 6, Figure 7, Figure 8).

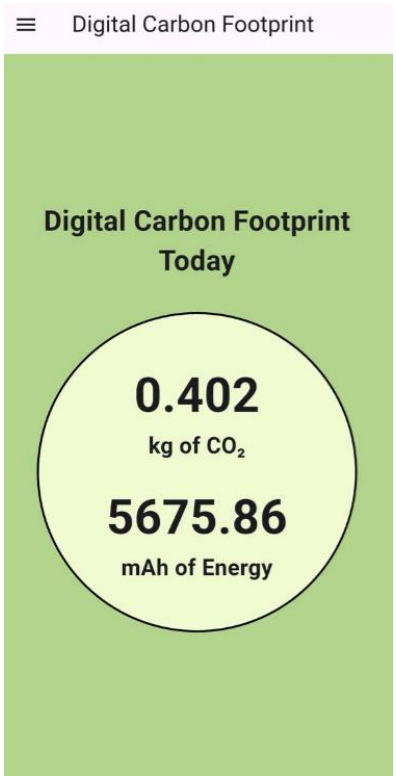

Figure 6: Test data 1

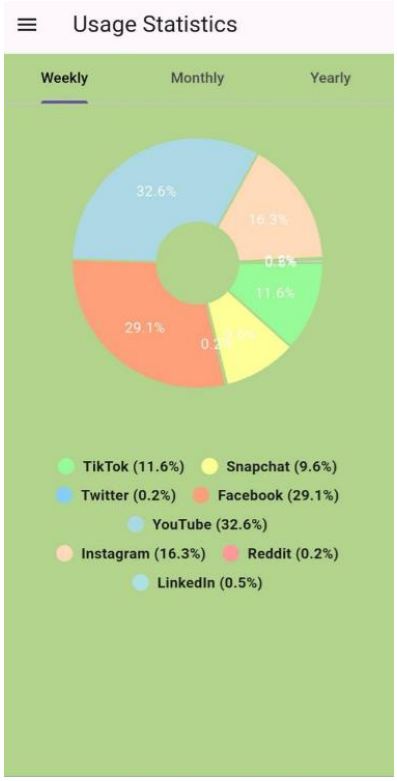

Figure 7: Test data 2

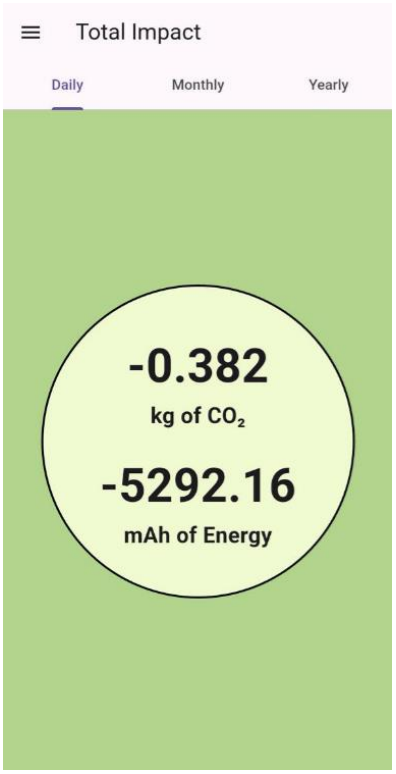

Figure 8: Test data 3

The correctness of the data cannot be assured for smaller applications as the existing data on digital carbon footprint is limited.

3.7 Limitations

The methodology is constrained by its reliance on emissions data derived entirely from Greenspector's 2021 report, which, while comprehensive, may not fully account for variations in device energy efficiency, regional energy grids, or broader digital habits. To enhance the accuracy and reliability of the app's calculations, further cross-referencing with additional data sources is essential. This would help verify the correctness of emission factors and provide a more nuanced understanding of the environmental impacts of various applications.

The application calculates CO2 emissions per minute. CO2 and mAh calculations can be seen in Figure 9.

```
/// CO₂ emissions (grams per minute) for each app
final Map<String, double> co2PerMinute = {
  'com.google.android.youtube': 0.46,
  'tv.twitch.android.app': 0.55,
  'com.twitter.android': 0.60,
  'com.linkedin.android': 0.71,
  'com.facebook.katana': 0.79,
  'com.snapchat.android': 0.87,
  'com.instagram.android': 1.05,
  'com.pinterest': 1.30,
  'com.reddit.frontpage': 2.48,
  'com.zhiliaoapp.musically': 2.63,
};

/// Energy consumption (mAh per minute) for each app
final Map<String, double> energyPerMinute = {
  'com.google.android.youtube': 8.58,
  'tv.twitch.android.app': 9.05,
  'com.twitter.android': 10.28,
  'com.linkedin.android': 8.92,
  'com.facebook.katana': 12.36,
  'com.snapchat.android': 11.48,
  'com.instagram.android': 8.90,
  'com.pinterest': 10.83,
  'com.reddit.frontpage': 11.04,
  'com.zhiliaoapp.musically': 15.81,
};
```

Figure 9: CO2 and mAh calculation

Currently, the app focuses on commonly used applications, potentially excluding niche apps due to a lack of available data on their environmental impact. Moreover, certain digital habits, such as the use of VPNs, are not accounted for in the current methodology, despite their potential to significantly influence energy consumption and emissions. Research is needed to establish mathematical formulas or standardized emission metrics for such practices before they can be integrated into the app's calculations. Additional studies and data collection are crucial to ensure the app provides an accurate and holistic representation of users' digital carbon footprints.

4. Methodology

The application development followed Design Thinking and User-Centred Design (UCD). UCD ensures usability, engagement, and accessibility.

4.1 User-Centred Design Process

The concept idea was cultivated in a dynamic environment at the Climathon Lappeenranta 2024. The concept owners got continuous feedback from other participants as well as the event organizers. The event participants' reception of the concept idea showed that they believed the application has a

potential for raising awareness on digital carbon footprint and influencing more sustainable behaviour if its scope was focused, and users would be influenced through positive reinforcement.

The idea cultivation process at the Climathon led to simplicity and clear core features being highlighted in the application design from the beginning. Once the application development commenced after the Climathon event, the findings originating from the idea cultivation shaped design requirements. Intuitive visualization, gamification, and actionable insights were emphasized.

The initial version of the application was developed using design thinking techniques which led to an interactive prototype. At the current stage only limited test data has been collected. Further user testing will provide insights for refinement.

Key constraints included data accuracy and privacy considerations. The application collects user app usage data, which cannot be done in iOS environment, and for this reason the first prototype is developed only for Android. Further development must address this issue.

5. Results and Impact

While formal testing results are not yet available, initial evaluations based on the response to the concept idea at Climathon Lappeenranta 2024 indicate the potential for the application to significantly impact user behaviour and awareness.

Participants at the event expressed strong interest in the concept of a digital carbon footprint, demonstrating both curiosity and a willingness to learn more about how their digital habits contribute to environmental impact. This enthusiasm suggests a readiness to adapt their behaviour to be more sustainable, provided the right tools and information are made accessible.

- **User Engagement:** Early feedback highlights the app’s intuitive interface and the clear visualization of the user’s digital carbon footprint as significant strengths. These elements make it easier for users to engage with the data and understand their impact.
- **Behavioural Change:** The app's focus on real-time feedback, gamification, and actionable insights is expected to help users reduce screen time and adopt more sustainable digital habits in a way that feels achievable and rewarding.
- **Emission Awareness:** By offering relatable comparisons—such as equating emissions to physical activities like kilometres travelled or kilograms of beef produced—the app effectively bridges the gap between abstract environmental data and tangible, everyday behaviours.

The key goal of the application is to close the gap between users and actionable information, empowering them to implement changes in their everyday lives. Through gamified elements, the app ensures that adopting sustainable digital habits is both educational and engaging. Future testing with a larger audience will provide quantitative metrics on user engagement, behavioural changes, and emissions reductions.

6. Discussion

This project highlights the feasibility and value of cultivating innovative ideas in a collaborative, think-tank-like setting, such as the Climathon, and further refining the selected concept into a tangible product.

The dynamic, social environment of the Climathon, where like-minded individuals work together in groups to address a specific challenge, fosters creativity and problem-solving. This collaborative process not only generates diverse solutions but also adds inherent value that translates directly into the development of practical outcomes—such as this mobile application.

The resulting app effectively raises awareness about digital carbon footprints and encourages sustainable habits. By combining intuitive design with educational features, it offers a user-friendly platform that empowers individuals to learn about and reduce the environmental impact of their digital activities. This seamless progression from ideation to implementation demonstrates the potential of participatory innovation to drive meaningful change in addressing environmental challenges.

6.1 Key Contributions

Building on this collaborative foundation, the application delivers key contributions that enhance its effectiveness as a tool for driving sustainability:

- **Personalized Feedback:** Providing users with real-time emissions data and relatable comparisons encourages informed decision-making.
- **Behavioural Insights:** The app's simple, focused design ensures that users are not overwhelmed, which increases the likelihood of sustained behavioural change.
- **Research Potential:** Beyond its role as an educational tool, the application has the potential to serve as a valuable resource for further research into digital carbon footprints. By collecting anonymized user data on digital habits and emissions over time, the app could contribute to the body of scientific literature, offering insights into user behaviours and their environmental impact on a broader scale.

These contributions demonstrate the app's potential to bridge the gap between environmental awareness and action, empower users to make impactful changes in their everyday lives, and advance the understanding of digital sustainability through data-driven research.

6.2 Future Improvements

Several enhancements could further expand the app's functionality and impact:

- **Accessibility Enhancements:** Adding support for multiple languages, customizable color schemes, and assistive technologies like text-to-speech would make the app inclusive for a wider audience.
- **Additional Features:** Implementing push-notifications, goal emission level, and tangible data comparison for the user's digital carbon footprint would improve the application's overall impact.
- **Advanced AI Features:** AI-based recommendations could analyse user habits and suggest optimal behaviours to minimize emissions while maintaining productivity. This must be

implemented in a way that keeps the carbon emission tracker application's own emissions on a as low level as possible.

Further development and potential publishing of the application must be first agreed with the concept owners.

7. Conclusion and Future Work

The mobile application developed based on the concept cultivated at Climathon Lappeenranta 2024 serves as an innovative tool for raising awareness of digital carbon emissions and encouraging sustainable digital habits. While the app successfully bridges the gap between users and actionable information, its reliance on generalized emissions data highlights the need for more comprehensive research. Future work should focus on cross-referencing multiple data sources to ensure the accuracy and validity of emissions calculations, as well as expanding the dataset to account for variations in device energy efficiency, regional energy grids, and broader digital behaviours such as VPN usage.

Scaling the application to a broader audience is a key next step, accompanied by the integration of more advanced AI features to personalize feedback and predict optimal digital behaviours for users. These advancements, combined with continued research into digital carbon footprints, will not only enhance the app's reliability and functionality but also contribute to the scientific understanding of the environmental impact of digital activities. By addressing these areas, the application has the potential to become an important tool for education, behaviour change, and further research into digital sustainability.

8. Acknowledgments

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